



Nonequilibrium grain boundaries and their relaxation under oscillating stresses in columnar nickel nanocrystals studied by molecular dynamics

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ABSTRACT

Atomic structure of columnar nickel nanocrystals with $[112]$ column axis having nonequilibrium grain boundaries (GBs) containing extrinsic grain boundary dislocations (EGBDs) and its evolution under oscillating stresses are studied by molecular dynamics method. Energy of GBs as a function of the degree of nonequilibrium is evaluated. It is found that under loading by symmetrically oscillating stresses the nonequilibrium GBs generate lattice dislocations, which travel across the grains and are absorbed by opposite GBs thus resulting in a relaxation of the structure, long-range stress fields and the energy of GBs.

1. Introduction

One of important processes occurring during cold plastic deformation of polycrystalline materials is the accumulation of dislocations in grain boundaries (GBs), which act as strong barriers for slip [1–7]. This process results in the formation of extrinsic grain boundary dislocation (EGBD) arrays, which induce long-range stress fields and cause an enhanced latent energy [7,8]. GBs containing EGBDs are commonly called nonequilibrium GBs [3,7–9]. The nonequilibrium structure of GBs can significantly change their diffusion coefficient [10–14], the yield stress of polycrystals [15] and other properties.

Nonequilibrium GBs have been shown to be typical elements of the structure of bulk nanostructured materials processed by severe plastic deformation (SPD) methods [16–21]. Rybin and co-workers [22,23] studied the structure of materials after large plastic strains and developed the concept of mesodefects, which accumulate in GBs due to the trapped dislocations. They found that these defects were not only the consequence of plastic deformation but also determined the basic mechanism of grain subdivision during this process. Namely, misorientation mismatches at triple junctions due to EGBDs result in the formation of junction disclinations, whose relaxation occurs by the formation of new grain boundaries. Basing on the concepts developed by Rybin, earlier we proposed dislocation and disclination models of nonequilibrium GB structure in UFG materials [18–21]. These models allowed us to estimate the internal stress fields and excess energies in these materials and qualitatively explain their mechanical and diffusion properties [24,25].

The nonequilibrium GB structure is not stable and at elevated

temperatures relaxes resulting in the diminishing of internal stresses and excess energy [16,17]. Theoretical models of this process of grain boundary recovery process were proposed in [26–28].

Annealing is a very common but not the only treatment that leads to a relaxation of nonequilibrium defect structures in deformed materials. Recently, it has been shown that ultrasonic treatment (UST) can result in a significant recovery of GBs in UFG metals processed by SPD [29–36]. After UST with moderate amplitudes of oscillating stress (around 50 to 100 MPa) the microstructure of UFG nickel is characterized by better defined, straight GBs that is an indication on their equilibrium structure. Microstrains measured by X-ray diffraction and excess energy release measured by differential scanning calorimetric method decrease after UST with certain amplitudes. The relaxation of nonequilibrium GB structure under ultrasound results in an increase of the thermal stability of microstructure and enhancement of ductility of UFG nickel without a loss in the strength achieved due to SPD.

The phenomenon of ultrasound-assisted structure recovery has been addressed by two-dimensional dislocation dynamics simulations [34–36], which have shown that in a polycrystal containing triple junction disclinations and randomly distributed lattice dislocations exposed to the action of oscillating stresses the GBs absorb dislocations in a way, which results in a cancellation of the junction disclinations, i.e. GB recovery through GB-lattice dislocation interaction is possible. However, in UFG materials lattice dislocations can be not available in quantities enough for a compensation of the GB defects and one should think of possible sources of dislocations.

Direct studies of the structure, energy and kinetics of nonequilibrium GBs, their mechanical properties and behavior under

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different influences are possible by molecular dynamics simulations. A few kinds of a nonequilibrium GB structure in bicrystals and nanocrystals have been studied atomistically in several works [37–39]. In these works, the nonequilibrium state was created by introducing either random local shifts of GB atoms from their equilibrium positions [37] or free volumes by removing some GB atoms [38], or in the process of a construction of initial structures [37,39]. However, these kinds of nonequilibrium are essentially different from that caused by EGBDs, since they induce only short range internal stresses, whereas the EGBDs result in long-range stresses. While the short range stresses can relax by a local reconstruction of the GB atomic structure involving local diffusion or athermal displacements of atoms comparable to interatomic distances, those induced by EGBDs can relax only by their elimination from GBs requiring diffusion to the distances comparable to the grain size [26–28] or athermal processes of dislocation emission.

Quite recently, we have carried out molecular dynamics simulations of the behavior under oscillating stresses of a nonequilibrium GB containing a disclination dipole in bicrystalline nickel [40–42]. These simulations have shown that under a sinusoidal applied tension–compression stress the GB emits lattice dislocations, which travel across the grains and sink at free surfaces of the bicrystal. When the sign of the applied stress changes, these dislocations do not come back to the GB, since the dislocations are generated under the combined effect of internal stress field of the disclination dipole and applied stress, when the latter acts on the same direction. That is, an asymmetric process of lattice dislocation generation and GB recovery occurs under a symmetrically oscillating stress.

In the cited works, the free surfaces of the bicrystal were ideal sinks for dislocations emitted from the GBs and it is not clear, if this mechanism of GB recovery will be operating in a bulk polycrystal. Therefore, the aim of the present paper is to demonstrate that the unilateral process of nonequilibrium GB structure relaxation by lattice dislocation emission is a common mechanism of recovery in plastically deformed metals. For this, we develop a model nanocrystal of nickel, selected GBs of which contain EGBDs introduced by a special construction method. For different degrees of nonequilibrium, i.e. quantities of EGBDs, we determine the structures and excess energies of nanocrystals at temperatures 0 and 300 K and study their evolution during the action of an oscillating uniaxial tension–compression stress.

2. Computational details

2.1. Initial system construction

A columnar nickel nanocrystal having the column axis $[1\ 1\ 2]$ will be used for simulations. This system is very convenient to analyze the interaction between lattice dislocations and GBs, since it allows the slip of only edge dislocations with a Burgers vector $\frac{1}{2}\langle 110 \rangle$ and lines parallel to $\langle 1\ 1\ 2 \rangle$ on a slip plane $\{1\ 1\ 1\}$. Bicrystals with $\langle 1\ 1\ 2 \rangle$ tilt GBs have been extensively studied, for example, by Shimokawa et al. [43].

A computational cell containing a model nanocrystal is schematically presented in Fig. 1(a). The cell contains four grains, which have a common crystallographic axis $[1\ 1\ 2]$, the direction of which coincides with that of the z -axis. The grains have a regular hexagon-shape cross section with an edge a . Maximum size of the hexagons, $d = 2a$, will be taken as the grain size. Grains 1 and 2 have orientations of slip planes parallel to xOz and yOz planes, respectively. Taking the orientation of grain 1 as a reference, grains 3 and 4 are rotated to angles $\pm \alpha = \pm 20^\circ$ to form a symmetric high-angle tilt GB between them.

In order to construct a nanocrystal with equilibrium GBs, the grains were directly filled in by atoms belonging to properly oriented crystal lattices. The system thus obtained was checked to find the pairs of too close atoms in GB regions. In these pairs, one of the atoms was eliminated to avoid too strong repulsive interactions during the start of simulations.

To initialize the structures with nonequilibrium GBs, the

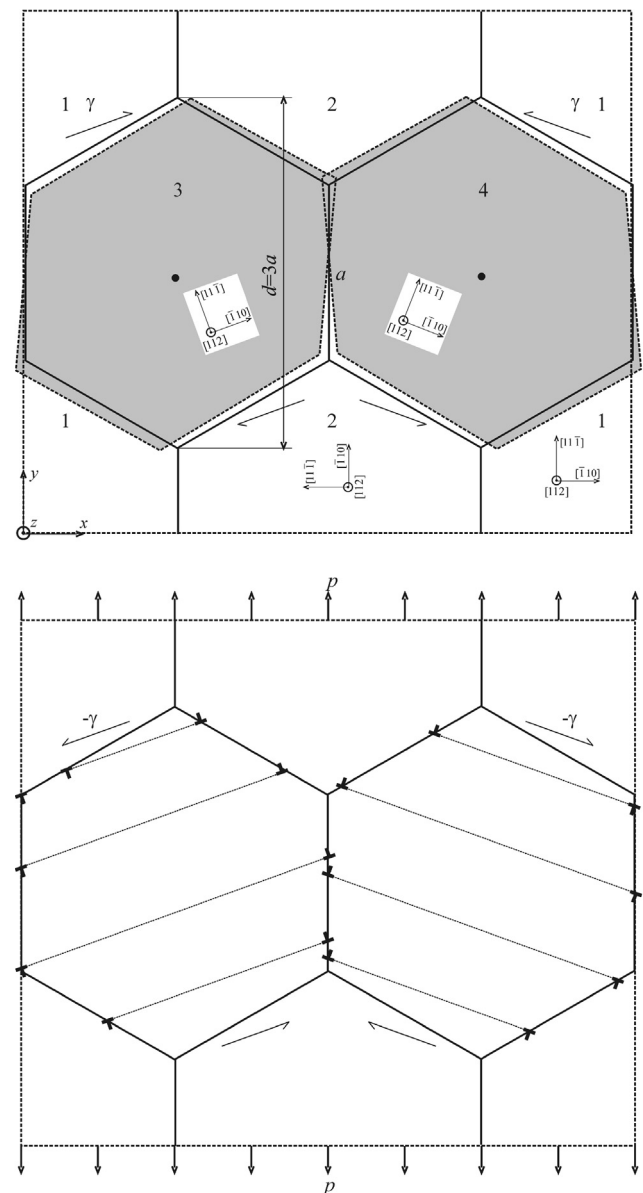


Fig. 1. Construction of the computation cell for columnar f.c.c. nanocrystals with a $[1\ 1\ 2]$ axis and nonequilibrium GBs.

construction method proposed in [44] is used. This method is based on shear strain of selected grains of the nanocrystal in order to introduce EGBDs into its boundaries. It is done as follows.

Assume that grains 3 and 4 of the model under consideration are free. Apply them simple shear strains γ in opposite directions by an imaginary slip of dislocations in $(1\ 1\ 1)$ slip planes so that the grains are distorted into the regions highlighted by a grey color in Fig. 1(a). The dislocation slip retains the ideal atomic structure of the lattice thus the grains will consist of a perfect lattice. However, the slip is limited to the volumes of the original grains and results in a formation of dislocations at their borders. In order to form these dislocations, one should now apply simple shears of opposite signs to the perfect grains occupying the gray regions.

As a result of this procedure, EGBDs will form in the boundaries of the deformed grains as schematically shown in Fig. 1(b). However, the geometrical procedure alone will not immediately form EGBDs having localized cores and it will require a relaxation of the structure. If one starts an energy minimization without precautions, the pairs of opposite signed dislocations will be attracted to each other and immediately

annihilate. In order to prevent this, one can apply an appropriate stress to the computational cell or fix some atoms near the GBs [44]. For the present system, one can apply a tensile stress p along the y axis of the computation cell, which induces a shear stress clamping the dislocations to GBs and exerts no shear stresses on the slip planes of grains 1 and 2.

In order to obtain nanocrystal systems with different levels of nonequilibrium, three values of shear strain, $\gamma = 0.05, 0.10$ and 0.15 , were applied by the above described procedure.

2.2. Simulation procedures

Simulations were carried out for nickel using embedded atom potential *niu3* developed in [45]. Corresponding lattice parameter is equal to $a_0 = 0.352$ nm. The grain size of nanocrystals was set equal to $d = 2a = 15$ nm. The periods of the computation cell are equal to $H_x = 2\sqrt{3}a = 25.981$ nm, $H_y = 3a = 22.500$ nm, and $H_z = |3a_0/2[112]| = 1.293$ nm. Periodic boundary conditions with constant pressure were applied along all directions (zero along x and z and zero or p along y).

All systems constructed were first relaxed at zero absolute temperature. The nanocrystal with equilibrium GBs was relaxed without any constraints by 10,000 MD steps.

To relax the systems with EGBDs, two protocols were used. In the first one, which will be further referred to as Protocol 1, initially 2000 MD steps at $T = 0$ K were done under the tensile stress $p = 0.07$ GPa applied along the y -axis, then the value of the stress was decreased by one per cent each 100 steps and set to zero in 10,000 steps. Finally, 20,000 steps of relaxation were done at zero stress. In Protocol 2, first 1000 minimization steps under the same tensile stress were carried out under the constraint of fixed GB atoms. For this, all atoms with energies higher than -4.20 eV were selected. Visualization showed that all these atoms lied in GB regions. Then 1000 steps were done without these constraints under the applied stress. The subsequent steps of this protocol were the same as those of Protocol 1.

The above described minimization methods were applied to each of the initial structures with nonequilibrium GBs, so six systems were generated, which we will denote further 0.05_1, 0.05_2, 0.10_1, 0.10_2, 0.15_1, and 0.15_2.

In order to obtain equilibrated structures at $T = 300$ K, MD runs for 160 ps were applied to all these systems with the time step $\Delta t = 2$ fs. To simulate the effect of an oscillating stress, tension–compression stress along the y -axis was applied to the equilibrated systems at the same temperature of $T = 300$ K. The value of the stress was given by equation $p = p_0 \sin(2\pi k/40)$ ($k = 1, 2, \dots, 40$) and for each value of the stress 2000 steps of MD run were done. Thus, the period of oscillations was equal to $\tau = 160$ ps. Up to ten cycles of the stress oscillations were simulated.

The frequency of the stress oscillations, $\nu = 1/\tau \approx 6$ GHz, is far higher than the frequencies of ultrasonic treatment applied in practice (about 20 kHz) and belong to the hypersonic range rather than to the conventional ultrasonic one. The very high rate of deformation is quite general for MD simulations and is related to limitations of the method. However, qualitative results on mechanisms of deformation obtained by MD simulations are considered to be valid for materials underwent to strains with ordinary rates [46].

Configurations of the atomic systems were recorded to an XMOL file each 2000 steps, i.e. after relaxation at a given value of the applied stress, so the visualization of the structures allowed one to observe changes in the atomic structure under hypersonic action. After each cycle of deformation, the systems were subjected to very short, for 200 steps, quenching in order to eliminate random displacements of atoms due to thermal oscillations. These relaxed systems were used to plot atomic energy maps and to calculate the total energy of defects.

Simulations were carried out by molecular dynamics code XMD [47]. Atomic structures were visualized by visualization codes RasMol [48] and OVITO [49,50]. To visualize the processes of partial lattice dislocations generation and glide, common-neighbor analysis [51]

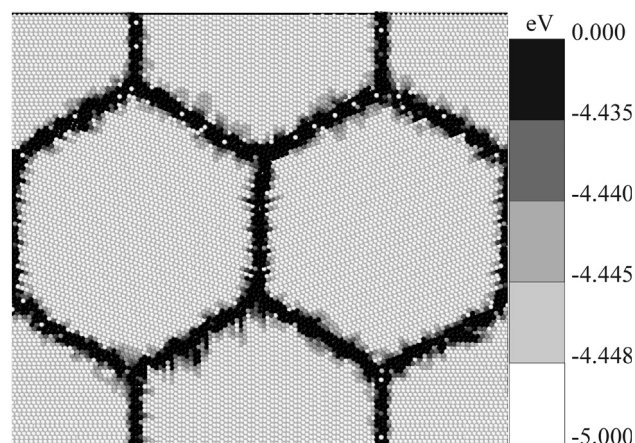


Fig. 2. Atomic energy map for the equilibrium structure of a nanocrystal with the grain size $d = 15$ nm at $T = 0$ K.

implemented in OVITO has been used.

3. Results

3.1. Equilibrium and nonequilibrium grain boundary structures and energies at zero absolute temperature

Atomic energy map of the nanocrystal with equilibrium GBs is presented in Fig. 2. As expected, this map is characterized by low energy of atoms inside grains and inhomogeneous distribution of high energies between GB atoms. This is typical for bicrystalline and polycrystalline materials with a perfect structure of GBs, which do not induce long-range internal stresses.

During energy minimization of all systems containing nonequilibrium GBs these GBs generate a few partial lattice dislocations, which glide across the grains or stop inside them, i.e. a partial relaxation of the nonequilibrium structure occurs (see [Supplementary movies Rel_0.05_1.mp4 to Rel_0.15_2.mp4](#)). The higher the level of nonequilibrium given by the value of shear strain γ , the more dislocations are generated. Dislocations at different positions are generated during relaxation of the same initial structures by the two protocols used, but their numbers and correspondingly the degrees of relaxation do not differ significantly.

Despite a partial relaxation, most of the EGBD content of GBs is retained on complete energy minimization. The high level of lattice distortions, which are caused by these EGBDs, can be visualized using Moiré patterns formed by an overlap of lattice images of nanocrystalline grains and the perfectly periodic pixel structure of a computer screen. Such a method of superposition of high-resolution electron microscopic images with an artificial printed lattice was used to visualize lattice distortions in nanocrystalline materials in Ref. [52]. The Moiré patterns obtained by visualizing the simulated atomic structures using RasMol at proper magnifications are presented in Fig. 3(a) and (b) for the cases of nanocrystals with initial structures 0.05_1 and 0.15_1. One can clearly see the distortions of lattice planes in the grains. As far as only a couple of partial dislocations are located inside the grains, all these distortions are induced by nonequilibrium GBs, i.e. are due to EGBDs.

Atomic energy maps for relaxed structures of all six systems are presented in Fig. 4e–f. The energy distributions for two systems obtained from the same initial structure by different protocols are quite similar. This means that the two ways of relaxation generate similar structures with similar energies. However, the higher the degree of nonequilibrium, the larger are differences. For the case of $\gamma = 0.15$ the second way of relaxation resulted in a small crack near one of the triple junctions. Evidently, the crack resulted in a partial relaxation of

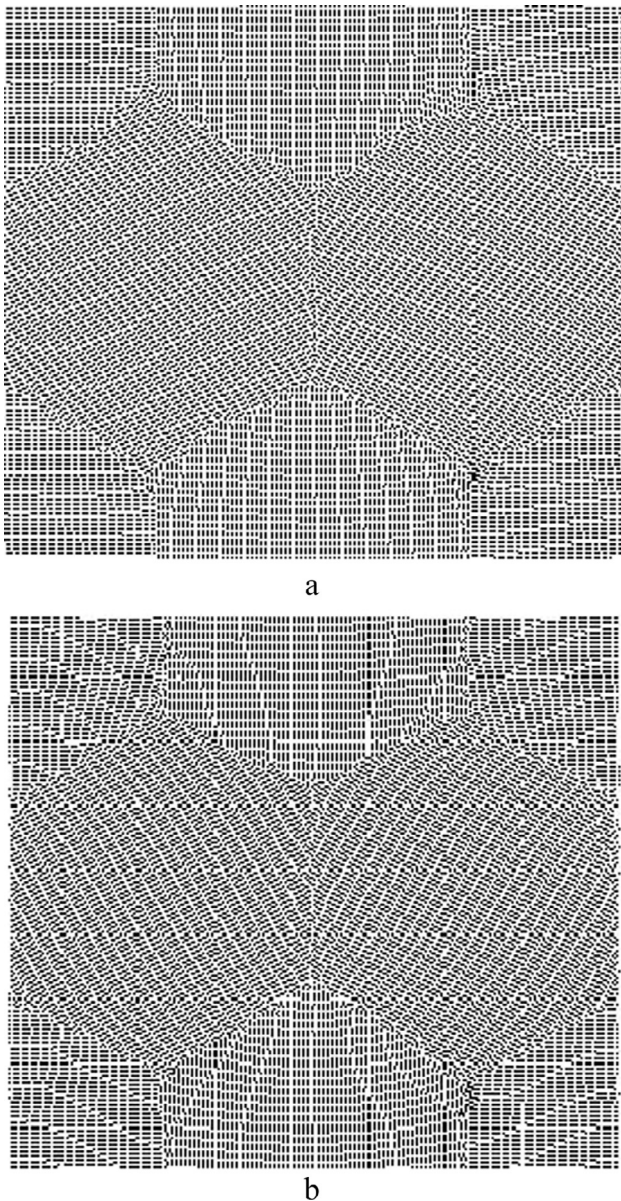


Fig. 3. Moiré patterns from the atomic structures of nanocrystals with EGBDs for initial states 0.05_1 (a) and 0.15_1 (b).

distortions of the lattice that is seen on the energy map.

Excess energies of atoms are directly related to the elastic stresses. Therefore, the energy maps considered also give evidence that inside the grains of the nonequilibrium nanocrystals irregular long-range stresses exist. A comparison of atomic maps for the three values of shear strain studied shows that the level of excess atomic energies and irregularities in their distribution increase with an increase of the nonequilibrium degree.

Presented in Table 1 are the values of average defect energy, E_{def} , in the nanocrystal with equilibrium GBs and those with nonequilibrium GBs relaxed by the two ways per unit area of GBs. The table also presents the values of the excess energy of nonequilibrium GBs over the energy of equilibrium GBs, E_{ex} .

As one can see from the table, Protocol 2 of energy minimization results in a lower energy of the systems. This means that it leads to a loss of more defect content of the nonequilibrium GBs during relaxation procedure. Nevertheless, in general the differences between the GB energies are not large and there is a general trend of an increase of the GB energy with the level of nonequilibrium.

3.2. Atomic structures at 300 K

Atomic energy maps for nanocrystals having nonequilibrium GB structures formed by shears $\gamma = 0.05, 0.10$, and 0.15 relaxed by Protocol 1 and then equilibrated at $T = 300$ K are presented in Fig. 5. Similar maps are obtained for nanocrystals relaxed by Protocol 2 too. The maps show that some relaxation of the long range stress fields in the grains occurred at room temperature. However, as one can see from Table 1, relaxation of only a small part of the GB excess energy occurs. This means that the GBs retained their EGBDs at this temperature. This should be expected on the basis of the fact that relaxation of such a kind of nonequilibrium GB structure can occur only by the climb of EGBDs, their escape to neighbor GBs and annihilation with EGBDs of opposite sign, i.e. it is a GB diffusion controlled process [26–28]. The rate of this relaxation is negligible at room temperature even in the time scale of seconds but not picoseconds. Therefore, one can be sure that the structures obtained by a relaxation in a couple of hundreds of picoseconds are in an equilibrated metastable structure, which cannot relax to EGBD free structures at $T = 300$ K.

3.3. Relaxation of nonequilibrium grain boundaries under cyclic loading

Consider now the behavior of nanocrystals at $T = 300$ K, when a cyclic tension–compression stress $p = p_0 \sin 2\pi \nu t$ is applied along the y -axis. Due to the specific crystallographic orientations of the grains of nanocrystals under study (see Fig. 1(a)), this stress will not induce shear stresses on the slip planes in grains No. 1 and 2, which are parallel or normal to the applied stress axis. On contrary, equal shear stresses on the slip planes of grains No. 3 and 4 are induced. When the applied stress is of a compression character, these shear stresses act in the same direction as the internal stresses from EGBDs, so an emission of lattice dislocations having the same signs as those of EGBDs will be facilitated. Vice versa, the tensile applied stress will clamp these dislocations to GBs or move back the dislocations emitted during the compression stage. At first thought, therefore, no effect on the nonequilibrium GB structure can be expected at a symmetrical cyclic loading.

However, the GBs are not only sources, but also good sinks for dislocations and this can introduce an asymmetry to this process.

Indeed, simulations show that at the amplitudes of the stress exceeding some critical value, nonequilibrium GBs emit partial lattice dislocations, which travel across the corresponding grains and are absorbed by opposite GBs. When the sign of the applied stress is changed, these dislocations are not emitted back from their new host boundaries, since the internal stress field acts against the applied one in this stage. For the particular nanocrystal under consideration emission of dislocations by GBs begins to occur at applied stress amplitudes approximately equal to $p_0 = 200$ MPa. The processes of dislocation emission during cyclic loading with amplitudes $p_0 = 200$ and 400 MPa are presented by Supplementary movies UST_200MPa_0.15_1.mp4 and UST_400MPa_0.15_1.mp4. An analysis of these records and the ones obtained at other values of the applied stress shows that at small amplitudes the events of lattice dislocation generation are rare and occur at larger number of cycles, while at high amplitudes several cycles are enough for an emission of all dislocations, which cancel the stress fields of EGBDs created in the initial structure.

Presented in Fig. 6 are the graphs of average potential energy of atoms in the nanocrystal with nonequilibrium GBs having the initial state 0.15_1 as a function of time during oscillating loading with three different values of the stress amplitude. From a comparison of these graphs, one can see that at lower amplitudes the energy of the system gradually decreases at a larger number of cycles and the energy levels off at a higher value than in the case of higher amplitudes. On contrary, at high amplitudes less number of cycles is needed for relaxation and a more complete relaxation is achieved.

Fig. 7 shows the change of the size of the same nanocrystal along the loading axis with time. Similarly to the energy, H_y also decreases during

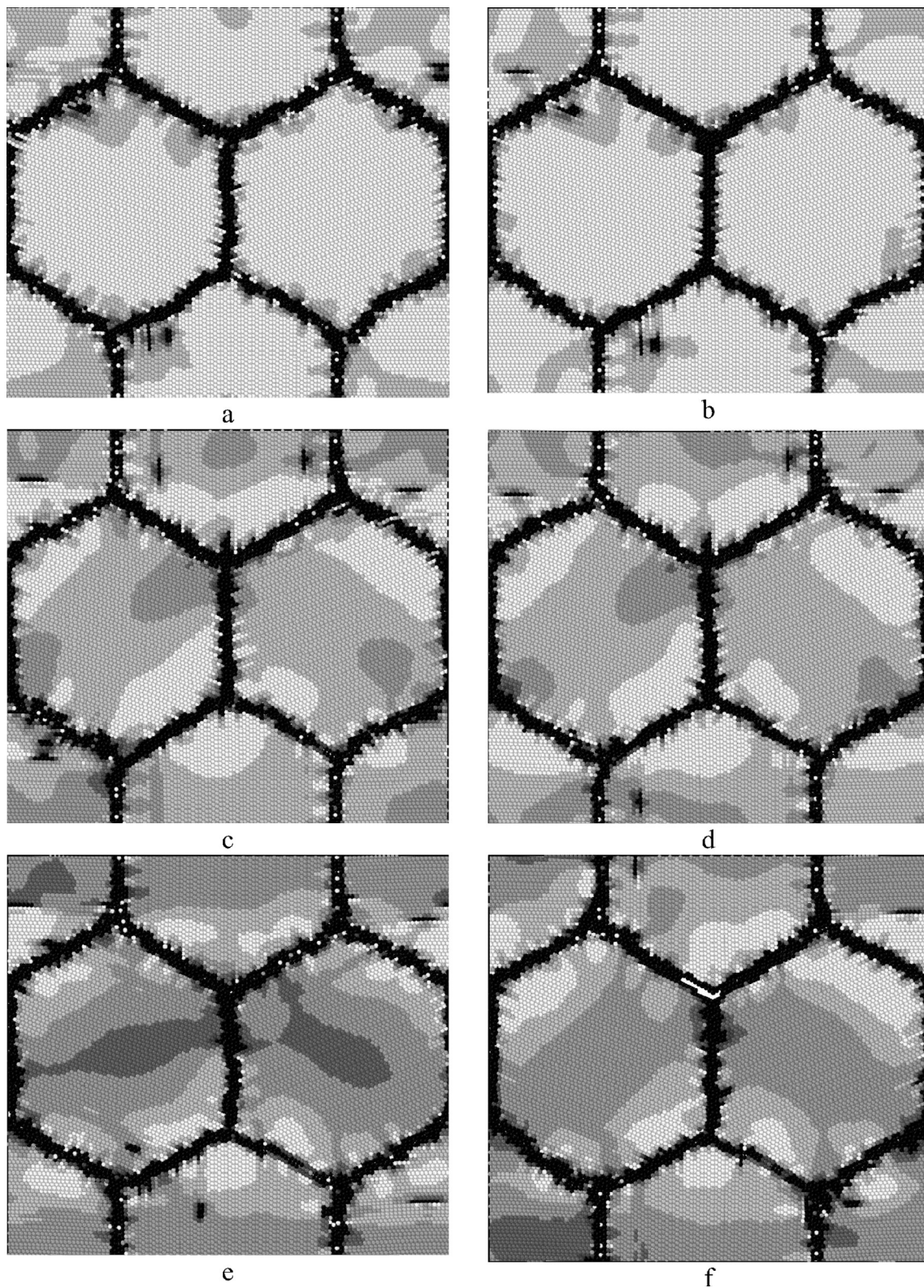


Fig. 4. Atomic energy maps for relaxed structures of nonequilibrium nanocrystals containing EGBDs formed by shear strains $\gamma = 0.05$ (a, b), 0.10 (c, d), and 0.15 (e, f) at $T = 0$ K obtained by protocols 1 (a, c, e) and 2 (b, d, f).

cyclic loading and the amount of this change increases with the stress amplitude. This is related to the fact that dislocations gliding across the grain cause plastic strain that results in some upsetting of the system in the vertical direction.

Atomic energy maps of the system 0.15_1 after 10 cycles of loading

with stress amplitudes $p_0 = 200, 300$ and 400 MPa are presented in Fig. 8(a)–(c).

As one can see from Fig. 6 and from a comparison of Fig. 8(a)–(c) with Fig. 5(c), at the stress amplitude 200 MPa cyclic straining results only in a little decrease of the energy and a slight redistribution of long

Table 1
Energies of grain boundaries in equilibrium and nonequilibrium nanocrystals at zero and room temperatures.

State No.	Equilibrium	0.05_1	0.05_2	0.10_1	0.10_2	0.15_1	0.15_2
$T = 0 \text{ K}$							
$E_{\text{def}}, \text{Jm}^{-2}$	1.27	1.41	1.39	1.56	1.54	1.79	1.74
$E_{\text{ex}}, \text{Jm}^{-2}$	0	0.14	0.12	0.29	0.27	0.52	0.47
$T = 300 \text{ K}$							
$E_{\text{def}}, \text{Jm}^{-2}$	1.26	1.36	1.34	1.52	1.50	1.75	1.64

range stresses in the grains. With an increase of the amplitude the relaxation of the stress fields becomes more pronounced and at $p_0 = 400 \text{ MPa}$ it becomes complete. In Fig. 8(c), one can see some small residual excess energies of atoms in the grains, which are caused by stacking faults due to the passage of partial lattice dislocations. Despite this excess energy in the grains, the total energy of defects in this state is even slightly less than in the state obtained by MD run from a state with equilibrium GBs (Table 2).

4. Discussion

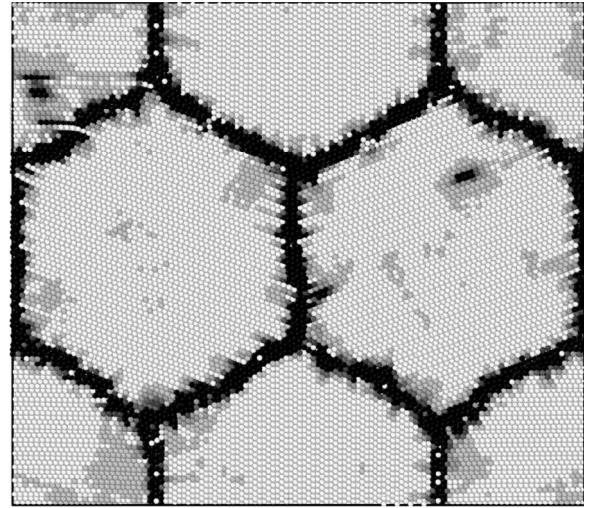
According to dislocation and disclination models of nonequilibrium GBs [9,18–21], the system of EGBDs depicted in Fig. 1 can be formally decomposed into systems of sessile and gliding EGBDs having Burgers vectors normal and tangential to the GB plane, respectively. The former are modeled by junction disclinations forming a quadrupole or a higher-order multipole configuration. Both the density of tangential EGBDs, ρ_t , and strength of disclinations, ω , are proportional to the shear strain of grains: $\rho_t, \omega \propto \gamma$. Energies of the two EGBD systems are proportional to the squares of ρ_t and ω , respectively. Therefore, the total energy of EGBD system quadratically depends on the shear strain, i.e. $E_{\text{ex}} \propto \gamma^2$.

For an approximate comparison, let us make a numerical estimate for the disclination component of the EGBD system. Due to shear strain γ of two adjacent grains 3 and 4 the GB of length a between them changes its misorientation angle by $\omega = 2\gamma \cos \alpha \approx 1.88\gamma$. This results in the formation of a disclination dipole with the strength equal to ω on each of the two vertical GBs in Fig. 1. Taking account for periodic boundary conditions, the resulting junction disclination system can be represented by an infinite disclination lattice, which is shown in Fig. 9. This system can be considered to consist of periodic disclination dipole walls (DDWs) of alternating signs, which are denoted by figures $0, \pm 1, \pm 2, \dots$. Each DDW consists of disclination dipoles with the strength ω , separation of disclinations a and period $h = 3a$. The spacings between neighbor DDWs are equal to $a\sqrt{3}$. The elastic energy of the system in the computation cell per unit length along z -axis can be calculated as follows:

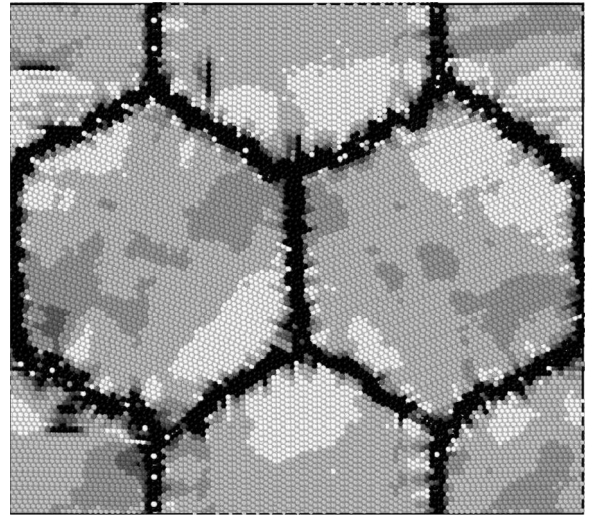
$$\frac{W}{L} = [(\gamma_0 + \gamma_1) + (\gamma_{0,1} + \gamma_{0,-1} + \gamma_{1,2}) + (\gamma_{0,-2} + \gamma_{0,2} + \gamma_{1,-1} + \gamma_{1,3}) + \dots] H_z, \quad (1)$$

where $\gamma_0 = \gamma_1$ are specific energies of each DDW per unit area and γ_{ij} are interaction energies of pairs of DDWs with numbers i and j . The round parentheses in Eq. (1) group self energies of DDWs, energies of nearest-neighbor, second-neighbor interactions etc. The self energy of a DDW was calculated by Shih and Li [53], while more general formulae for interaction energies of DDWs displaced with respect to each other along the axes x and y were obtained by Nazarov et al. [54,55] (see, for example Eqs. (2) and (3) in Ref. [55]). Adapting these formulae for the present system, one can obtain respectively for the self, first- and second-neighbor interaction energies:

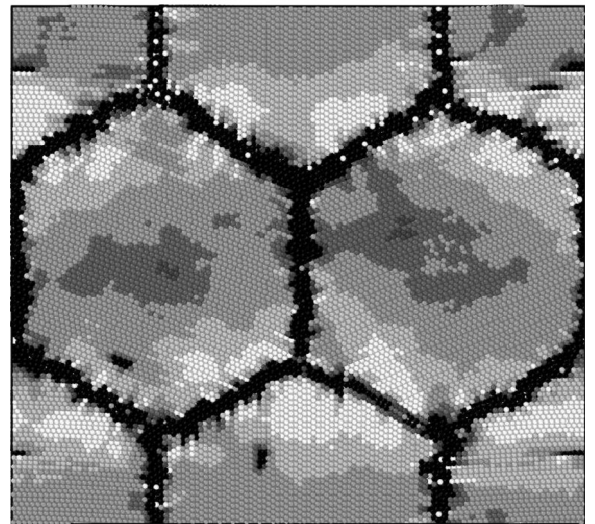
$$\gamma_0 = \frac{G\omega^2 H_z}{32\pi^3(1-\nu)} f(\lambda), \quad \gamma_{0,1} = -\frac{G\omega^2 H_z}{16\pi^3(1-\nu)} g(\delta, \lambda), \quad \gamma_{0,2} = \frac{G\omega^2 H_z}{16\pi^3(1-\nu)} g(2\delta, \lambda), \quad (2)$$



a



b



c

Fig. 5. Atomic energy maps of nonequilibrium nanocrystals containing EGBDs formed by shear strains $\gamma = 0.05$ (a), 0.10 (c), and 0.15 (e) relaxed by Protocol 1 and equilibrated at $T = 300 \text{ K}$.

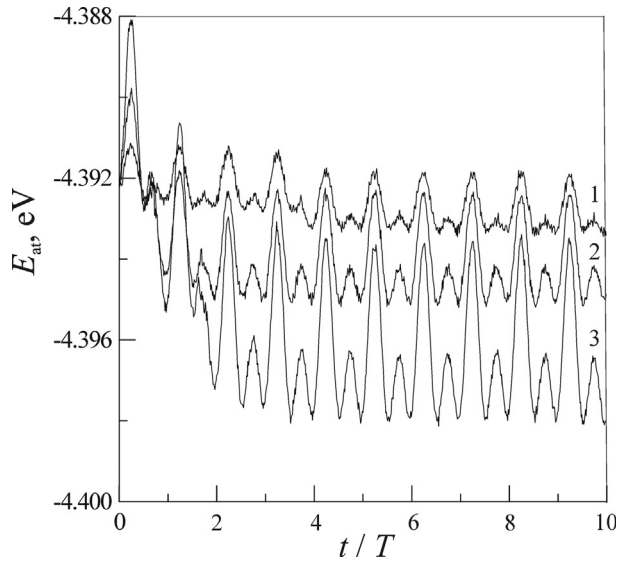


Fig. 6. Change of the average potential energy of atoms in the nanocrystal with nonequilibrium GBs having the initial state 0.15_1 with the number of tension–compression cycles at stress amplitudes $p_0 = 200$ (1), 300 (2), and 400 MPa (3) at temperature $T = 300$ K.

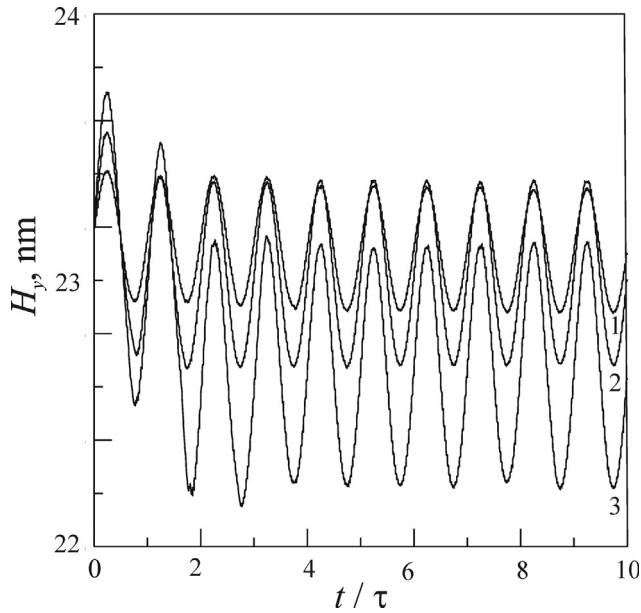


Fig. 7. Change of the size of the nanocrystal with nonequilibrium GBs having initial state 0.15_1 along the loading axis with the number of tension–compression cycles at stress amplitudes $p_0 = 200$ (1), 300 (2), and 400 MPa (3) at temperature $T = 300$ K.

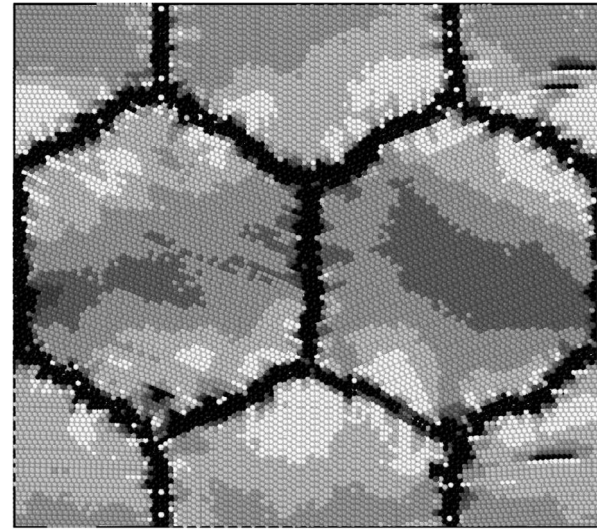
where $\lambda = \pi a/H_z = \pi/3$, $\delta = \pi a\sqrt{3}/H_z = \pi/\sqrt{3}$, and

$$f(\lambda) = 2 \int_0^{2\lambda} \sin u du \int_0^\infty \frac{v dv}{\cosh v - \cos u}, \quad (3)$$

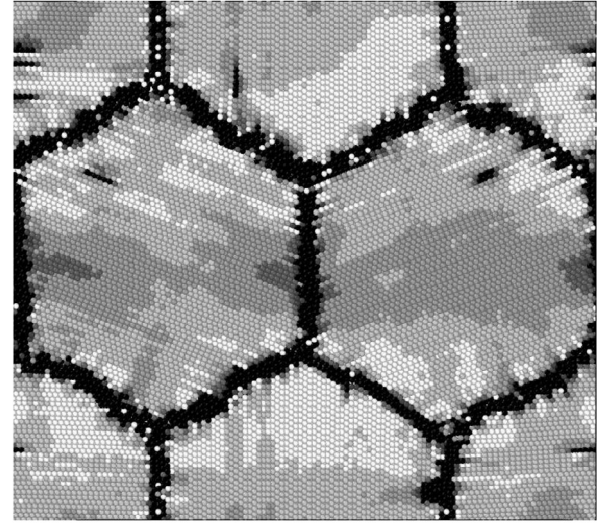
$$g(\alpha, \lambda) = f(\lambda) - 2 \int_0^{2\lambda} \sin u du \int_0^{2\alpha} \frac{v dv}{\cosh v - \cos u}. \quad (4)$$

The function $f(\lambda)$ and integral in Eq. (4) are calculated by an expansion to series. The latter series is converged well for $\alpha < 2.6$ [55].

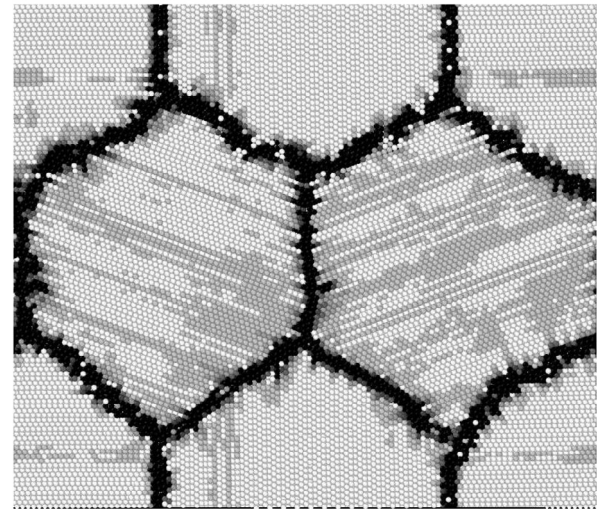
For the DDW system under consideration $f(\lambda) = 6.94$, $g(\delta, \lambda) = 0.74$, while the argument of $g(2\delta, \lambda)$ is well above the interval of convergence for the corresponding series. Nevertheless, a calculation for the edge of this interval yields $g(2.6, \lambda) = 0.19$, i.e. nearly four times less than $g(\delta, \lambda)$. Therefore, one can expect that $g(2\delta, \lambda)$ is nearly an order of magnitude



a



b



c

Fig. 8. Atomic energy maps of the system 0.15_1 after 10 cycles of loading with stress amplitudes $p_0 = 200$ (a), 300 (b), and 400 MPa (c) at temperature $T = 300$ K.

Table 2

Defect energies per unit GB area in nanocrystal 0.15_1 after 10 cycles of oscillating loading with different amplitudes of the stress.

p_0 , MPa	E_{def} , eV	E_{def} , Jm ⁻²
0	0.0183	1.75
200	0.0174	1.66
300	0.0158	1.50
400	0.0127	1.21

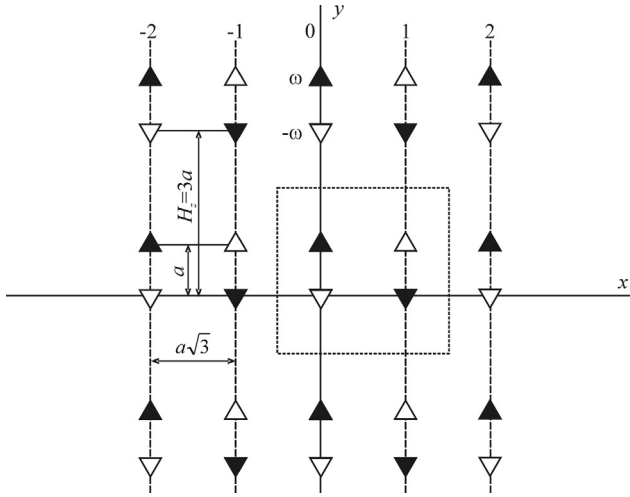


Fig. 9. Junction disclination array formed due to straining of the grains of a nanocrystal according to the scheme of Fig. 1. Dotted rectangle shows the borders of the computation cell.

less than $g(\delta, \lambda)$ and, therefore, one can take into account only the self and nearest-neighbor interaction energies of DDWs in Eq. (1).

Substituting the values of $f(\lambda)$ and $g(\delta, \lambda)$ into Eqs. (2) and (3) and taking into account the relation $H_z = 3a$, one finally obtains from Eq. (1)

$$\frac{W}{L} = \frac{0.08G\omega^2a^2}{1-\nu} \approx \frac{0.28Ga^2\gamma^2}{1-\nu}. \quad (5)$$

Further substituting the Poisson's ratio for Ni $\nu = 0.31$ and the grain boundary length $a = 7.5$ nm, dividing the elastic energy to the total length of GBs in computation cell, $12a$, one finally finds the excess GB energy in the nanocrystal,

$$\gamma_{ex} = \frac{0.024Ga\gamma^2}{1-\nu} \approx 20.3\gamma^2 [\text{Jm}^{-2}]. \quad (6)$$

Fig. 10 presents the graphs of shear strain dependencies of the GB excess energy obtained by the two protocols of MD relaxation and using Eq. (6). One can see a good agreement between the MD and elasticity theory calculations. Taking into account the other component of EGBD structure consisting of dislocations with tangential Burgers vectors, one can conclude that the total elastic energy will be higher than that calculated by Eq. (6). It is evident that at high values of γ the elastic energy will be higher than the excess energy calculated by MD. This is consistent with the fact that a partial relaxation of the nonequilibrium structure occurs during MD relaxation. On the other hand, atomistic calculations will still give higher values of excess energy at low values of γ . This means that in this case a significant part of the excess energy is associated with the core energy of EGBDs, which is not included in elasticity theory calculations.

For the maximum value of shear strain studied here the strength of disclinations is equal to $\omega \approx 16^\circ$. This value is near the upper limit of stability of disclinations for the given grain size. Indeed, in Ref. [56] it was shown that a negative wedge disclination in bicrystalline nickel

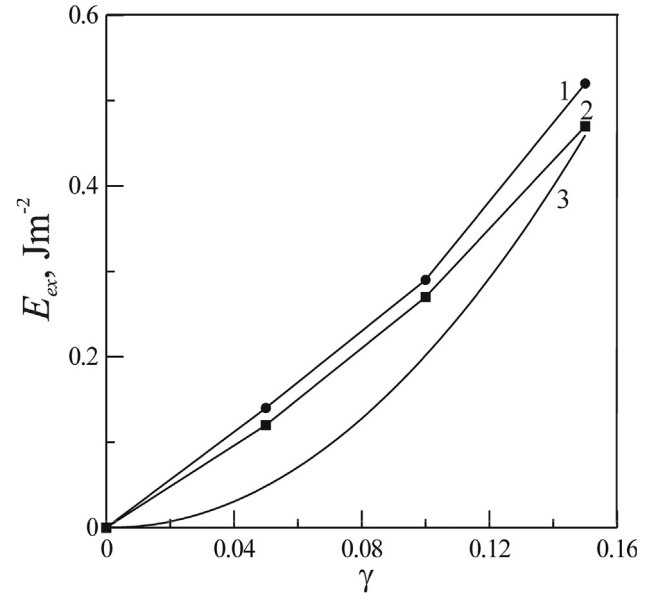


Fig. 10. Graphs of the dependence of GB excess energy on the shear strain of grains resulting in the formation of EGBDs plotted basing on the results of molecular dynamics simulations by protocols 1 (1) and 2 (2) and theoretical evaluations in a disclination model (3).

cylinder of radius R resulted in the formation of a crack at its core above a critical strength $\omega_c \approx 55^\circ/\sqrt{R}$, where R is substituted in nanometers. For screened disclination systems in a polycrystal, the grain size d is substituted in this equation and this gives for $d = 15$ nm $\omega_c \approx 14^\circ$, i.e. very close to 16° . As experimental studies of the microstructure of nanostructured materials processed by severe plastic deformation do not demonstrate the formation of cracks at triple junctions [16,17], EGBD densities in real UFG metals are below this limiting structure.

Thus, within the interval of EGBD densities, which can exist in real UFG metals processed by SPD, the present method of construction and MD relaxation of nanocrystals with dislocation containing GBs provides a good model of nanostructured materials with nonequilibrium GBs. This model can be further used to study the properties of nanostructured materials, for which atomistic models are necessary, for example, to calculate GB diffusion characteristics, kinetic properties, mechanisms of relaxation, mechanical properties etc.

The most common relaxation mechanism of the nonequilibrium GB structure involves the GB diffusion controlled climb of EGBDs [26–28] and requires elevated temperatures. Thus, most commonly the equilibrium GB structure is obtained by annealing of deformed materials.

However, relaxation of nonequilibrium GB structure can be achieved also by the action on materials using mechanical forces.

To authors' knowledge, Rupert and Schuh were the only authors who studied the response of the GB atomic structure, energy and mechanical properties of nanocrystals to cyclic straining [39]. They constructed nickel nanocrystals with the grain sizes from 3 to 10 nm by Voronoi tessellation method and simulated their behavior under the effect of annealing at 600 K and cyclic strains of loading and unloading. Both the thermal effect and straining resulted in a significant decrease of the average atomic energy, while the fraction of GB atoms did not change. This means that GB structure relaxation occurred without a change of the grain size. These relaxation processes are related to a change of the GB character and atomic rearrangements of the GBs. Along with this relaxation, an increase of the yield stress of the nanocrystals was observed.

Our work demonstrates that not only nonequilibrium GB structures of a non-dislocation character, but also the higher energy and long-living nonequilibrium structures such as EGBDs can relax due to cyclic straining in nanocrystals.

The relaxation effect of oscillating stresses on the structure of nonequilibrium GBs is consistent with the results of our recent experimental studies on ultrafine grained nickel processed by SPD methods such as high pressure torsion (HPT) and equal-channel angular pressing (ECAP). Transmission electron microscopic studies of these materials show that after UST some GBs acquire a banded contrast typical of equilibrium GBs just similar to that after moderate annealing [29–33]. Also, a reduction of the root mean square elastic strain measured by X-ray diffraction method is observed. However, a different behavior of this characteristic as a function of the ultrasound amplitude has been found. In earlier studies [29,30] it was shown to decrease monotonically with the amplitude, while in later studies we discovered that it first increased, attained a maximum and then decreased after achieving some value of the amplitude [33]. It seems that this behavior depends also on the behavior of lattice dislocations in the grains of as-prepared UFG metals. The present studies show that relaxation by the emission of lattice dislocations must show a threshold character, since the generation of dislocations occurs only at sufficiently high stress amplitudes. Thus, they provide a key for understanding the experimentally observed relaxation of internal stresses in UFG materials at sufficiently high amplitude of ultrasound.

During our simulations dislocation emission was observed only at very high applied stresses and in a few periods of stress oscillations. This is a common behavior for such idealized systems and for short time intervals available for molecular dynamics simulations. In real ultrafine grained materials processed by SPD the grain size is considerably larger. Also, the grains can contain dislocations, either individual or grouped into different substructure elements. Due to the relatively large grain size, interaction with existing lattice dislocations, thermal activation processes, relaxation of nonequilibrium GBs under UST can be observed at reasonable amplitudes of the stress in reasonable time intervals. This is confirmed by experimental studies, which demonstrate the occurrence of the effect at stress amplitudes of the order of 100 MPa and time intervals of about 30 s [29–33].

5. Summary

In this study, in order to explore the effect of EGBDs on the atomic structure, energy and behavior under high frequency cyclic straining of GBs, we constructed a specific atomistic model of a f.c.c. nanocrystal with $\langle 112 \rangle$ columnar grains and tilt GBs. This model has two advantages. First, it allowed us to construct most conveniently the initial structures and study dislocation emission from GBs due to the fact that only one slip system is active in each grain. Second, this atomic system allowed us to compare the results of energy calculations with those obtained in disclination modeling based on straight wedge disclinations.

The results of the study can be summarized as follows.

1. The atomistic model based on the proposed method of construction describes well the structure, long-range stress fields and energy of non-equilibrium GBs containing EGBDs in the whole range of stability of these GBs, i.e. for defect densities lower than those at which cracks at triples junctions of grains can form.
2. At low density of EGBDs the excess energy of GBs calculated by molecular dynamics simulations is higher than that derived from disclination and dislocation modeling that can be caused by the contribution of the core energy of EGBDs. At high densities, on contrary, the atomistically calculated energies can be lower due to a partial relaxation of EGBDs during molecular dynamics relaxation.
3. A combined action of internal stresses due to EGBDs and applied oscillating tension–compression stress results in an asymmetric process of the generation of lattice dislocations by nonequilibrium GBs and their glide across grains. This process leads to an effective elimination of EGBDs from the GBs and relaxation of their structure, energy and long-range stress fields. This relaxation has a threshold

character: it occurs at sufficiently high amplitudes of the periodic stresses. The higher the stress amplitude, the shorter is the time of full relaxation.

4. The process of lattice dislocation generation by nonequilibrium GBs can be considered as one of the possible underlying mechanisms of the effect of ultrasonic treatment on the structure and properties of ultrafine grained materials processed by deformation methods.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.commatsci.2018.05.015>.

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